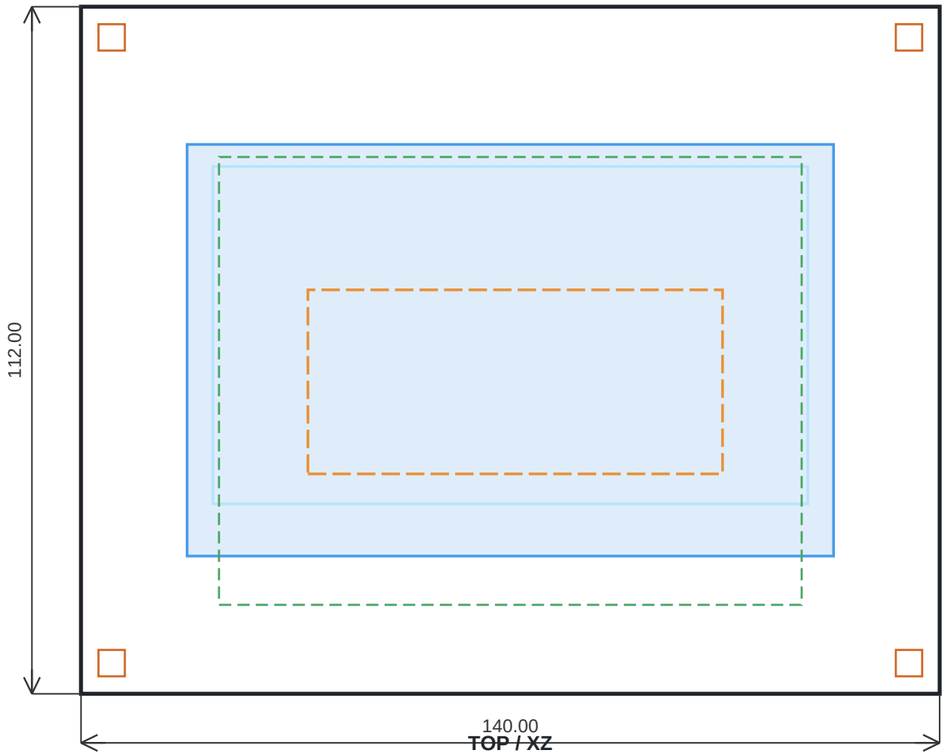
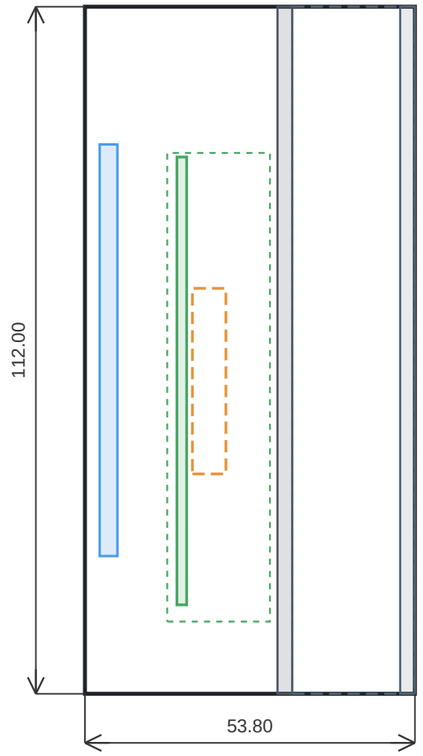
4.3-INCH COMPLETE ASSEMBLY - THREE VIEW

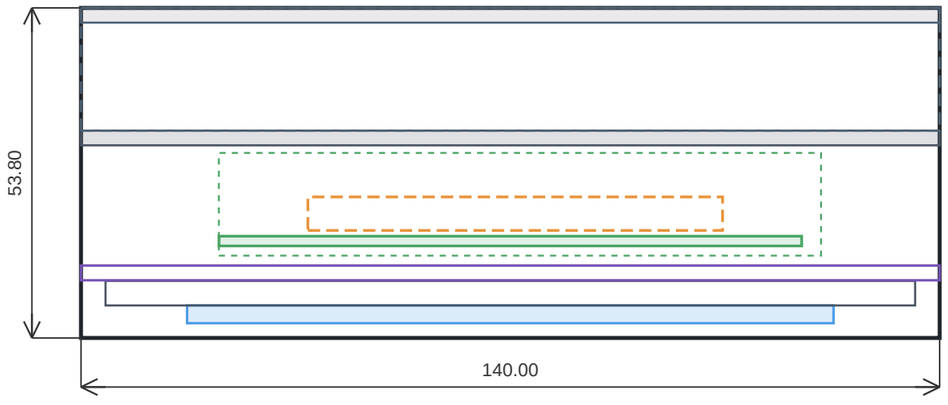
FRONT / XY



RIGHT / ZY



TOP / XZ



VIEW KEY

- Blue: exact official LCD envelope.
- Green solid: official 95 x 73 Dock PCB.
- Orange dashed: official installed Core envelope.
- Green short-dash: complete Dock STEP envelope.
- Purple: printed Dock tray.
- Gray: printed enclosure.

VERIFY BEFORE FINAL PRINT

- Orthographic geometry is drawn at true 1:1.
- The STEP envelope includes ports extending below PCB.

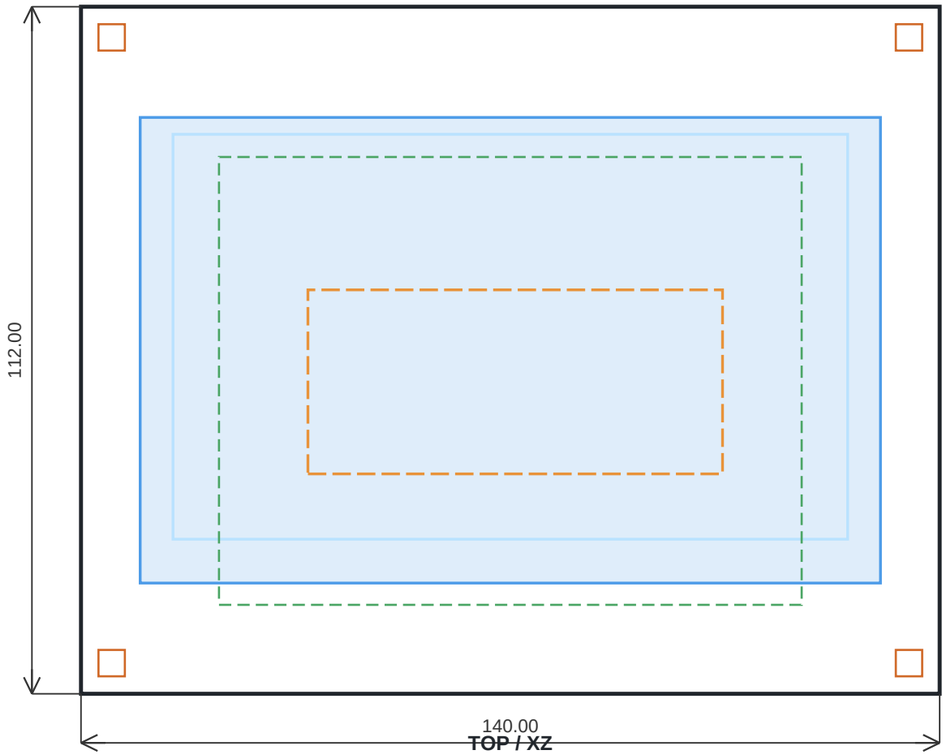
20 x 20

100 mm calibration bar

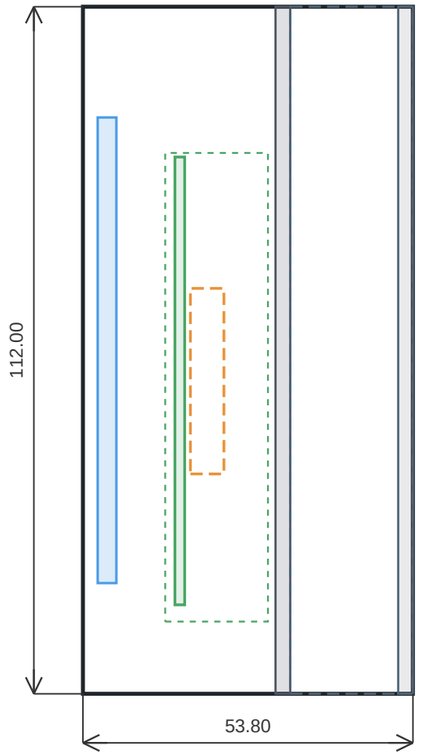
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

5.0-INCH COMPLETE ASSEMBLY - THREE VIEW

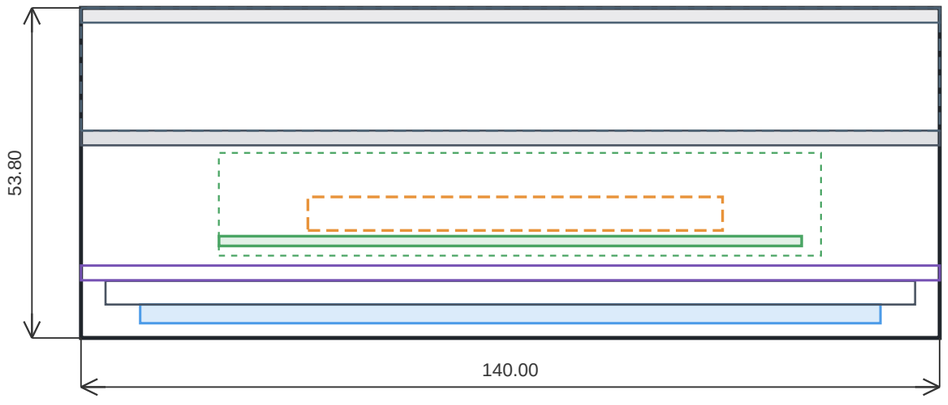
FRONT / XY



RIGHT / ZY



TOP / XZ



VIEW KEY

- Blue: exact official LCD envelope.
- Green solid: official 95 x 73 Dock PCB.
- Orange dashed: official installed Core envelope.
- Green short-dash: complete Dock STEP envelope.
- Purple: printed Dock tray.
- Gray: printed enclosure.

VERIFY BEFORE FINAL PRINT

- Orthographic geometry is drawn at true 1:1.
- The STEP envelope includes ports extending below PCB.

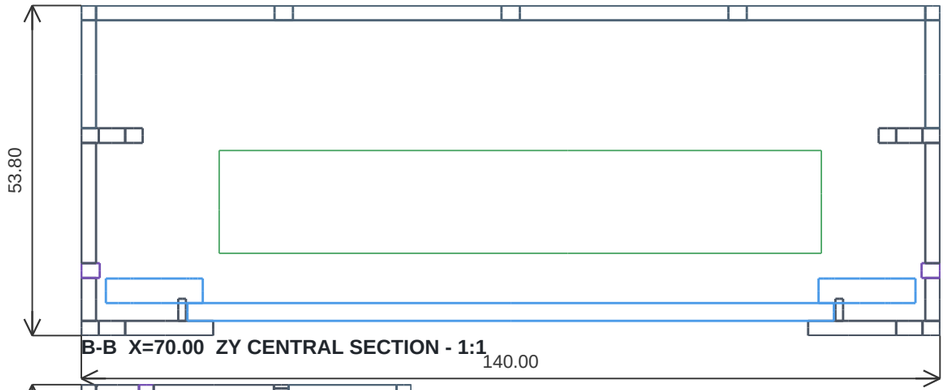
20 x 20

100 mm calibration bar

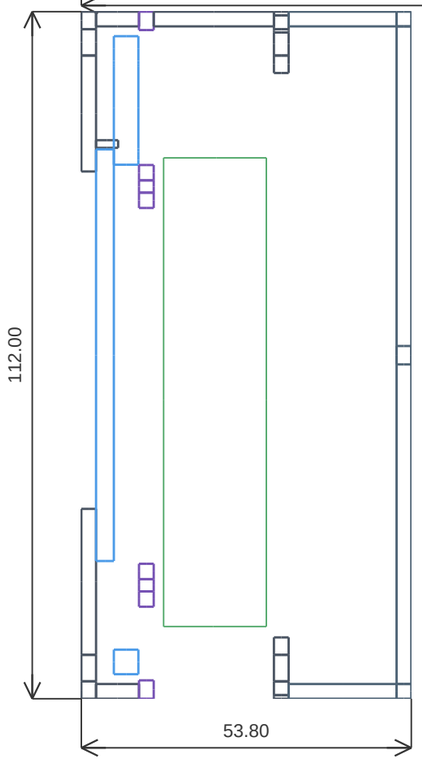
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

4.3-INCH COMPLETE ASSEMBLY - EXACT SECTIONS

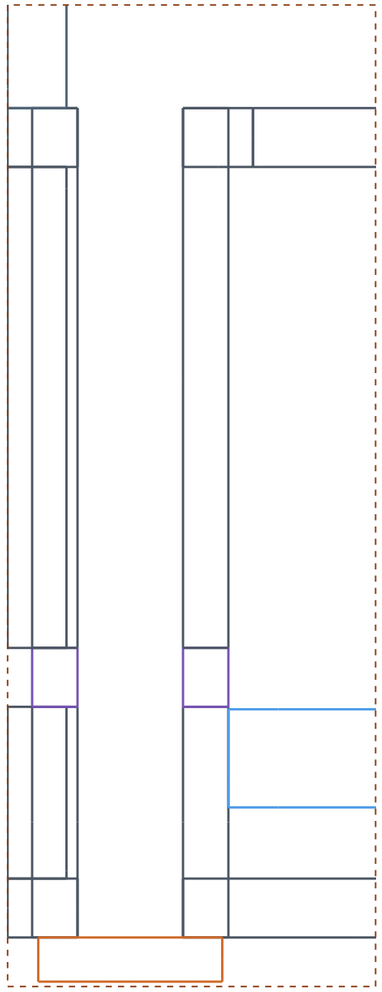
A-A Y=56.00 XZ CENTRAL SECTION - 1:1



B-B X=70.00 ZY CENTRAL SECTION - 1:1



C-C CASE SNAP PIN - 4:1



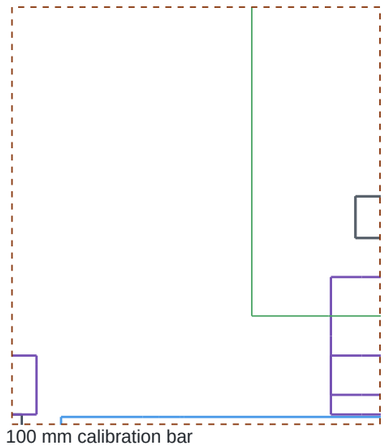
Z DATUMS FROM FRONT

- Front outer face: 0.00
- LCD: 2.40 to 5.30
- Retainer rear: 9.30
- Tray: 9.40 to 11.80
- Dock PCB: 15.00 to 16.59
- Core STEP: 17.523 to 22.992
- Dock+Core STEP: 13.415 to 30.165
- Access-frame inner face: 31.40
- Access-frame latch plane: 33.80
- Service-cap inner face: 51.40
- Service-cap outer face: 53.80
- Pin overall: 37.00; head front: -1.80

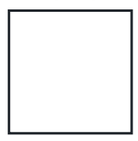
VERIFY BEFORE FINAL PRINT

- Minimum STEP-to-service-cap clearance: 21.235.
- Dock: 4 x M2.5 x 6 pan-head; printed square pilot 2.00 x 4.80 deep.
- Dock and Core are drawn as the overall envelope of the official STEP.
- Printed parts use their delivered STL facets.

D-D DOCK M2.5 SCREW BOSS - 4:1



20 x 20



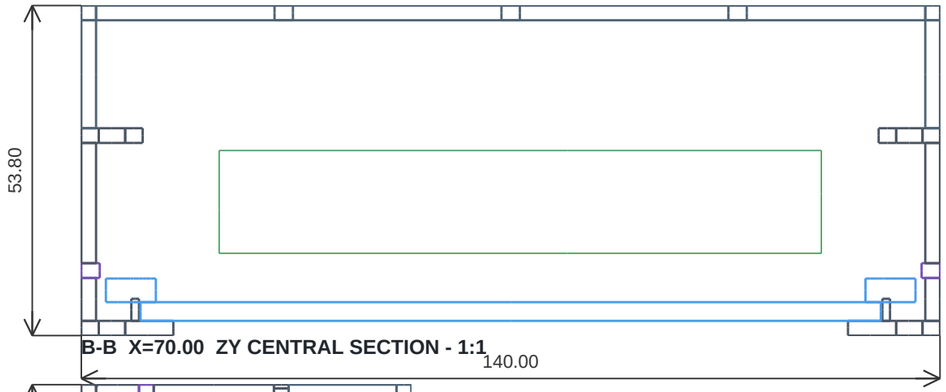
100 mm calibration bar



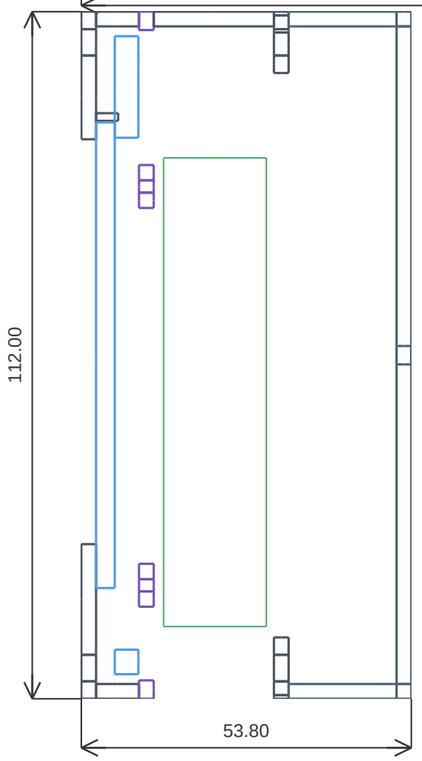
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

5.0-INCH COMPLETE ASSEMBLY - EXACT SECTIONS

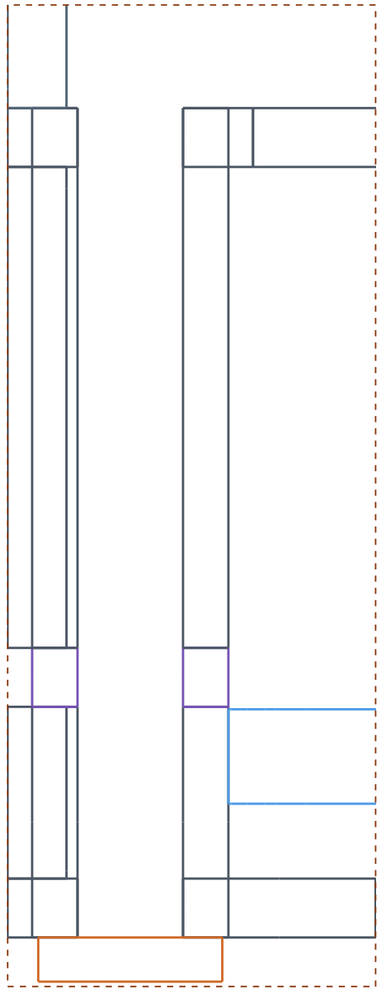
A-A Y=56.00 XZ CENTRAL SECTION - 1:1



B-B X=70.00 ZY CENTRAL SECTION - 1:1



C-C CASE SNAP PIN - 4:1



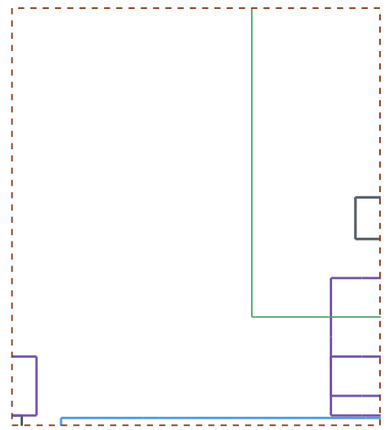
Z DATUMS FROM FRONT

- Front outer face: 0.00
- LCD: 2.40 to 5.45
- Retainer rear: 9.30
- Tray: 9.40 to 11.80
- Dock PCB: 15.00 to 16.59
- Core STEP: 17.523 to 22.992
- Dock+Core STEP: 13.415 to 30.165
- Access-frame inner face: 31.40
- Access-frame latch plane: 33.80
- Service-cap inner face: 51.40
- Service-cap outer face: 53.80
- Pin overall: 37.00; head front: -1.80

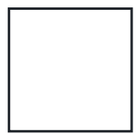
VERIFY BEFORE FINAL PRINT

- Minimum STEP-to-service-cap clearance: 21.235.
- Dock: 4 x M2.5 x 6 pan-head; printed square pilot 2.00 x 4.80 deep.
- Dock and Core are drawn as the overall envelope of the official STEP.
- Printed parts use their delivered STL facets.

D-D DOCK M2.5 SCREW BOSS - 4:1



20 x 20



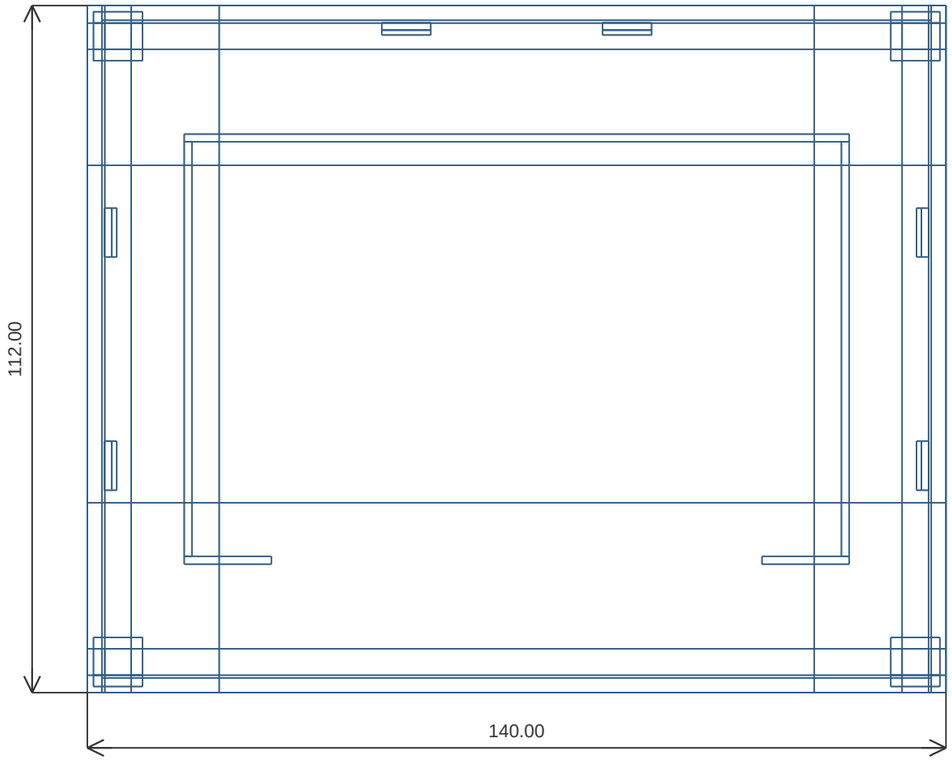
100 mm calibration bar



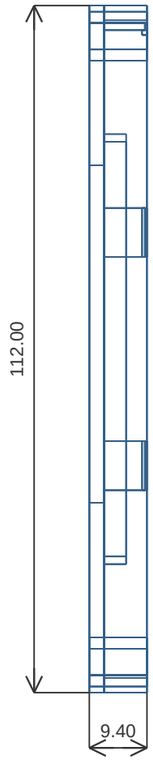
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - FRONT SHELL 4.3

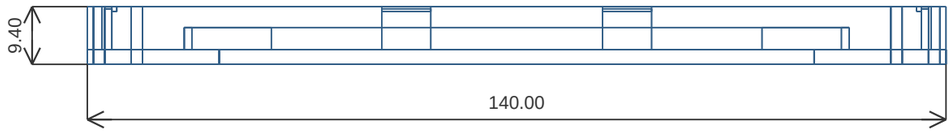
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



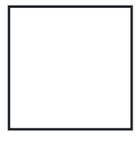
CONTROLLED GEOMETRY

- Source: front_shell_43_snap.stl
- STL bounds: 140.00 x 112.00 x 9.40
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: XY display face on bed; supports: none.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



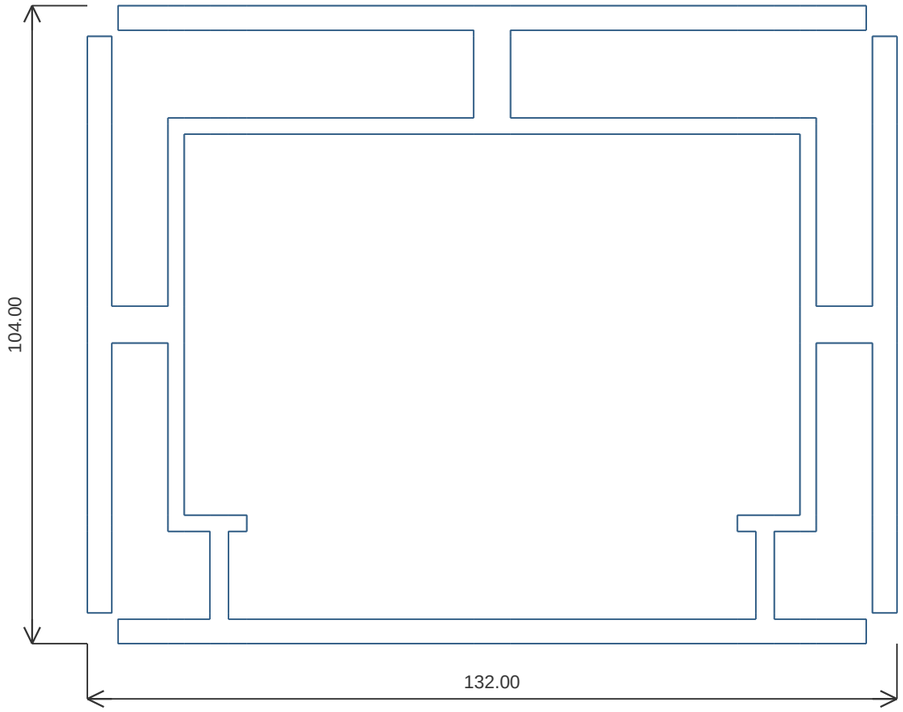
100 mm calibration bar



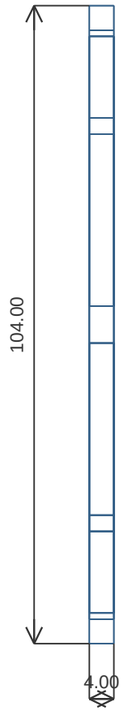
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - LCD RETAINER 4.3

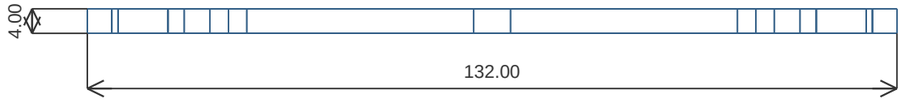
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



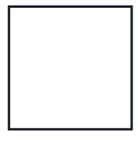
CONTROLLED GEOMETRY

- Source: lcd_retainer_43_snap.stl
- STL bounds: 132.00 x 104.00 x 4.00
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: XY flat face on bed; supports: none.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



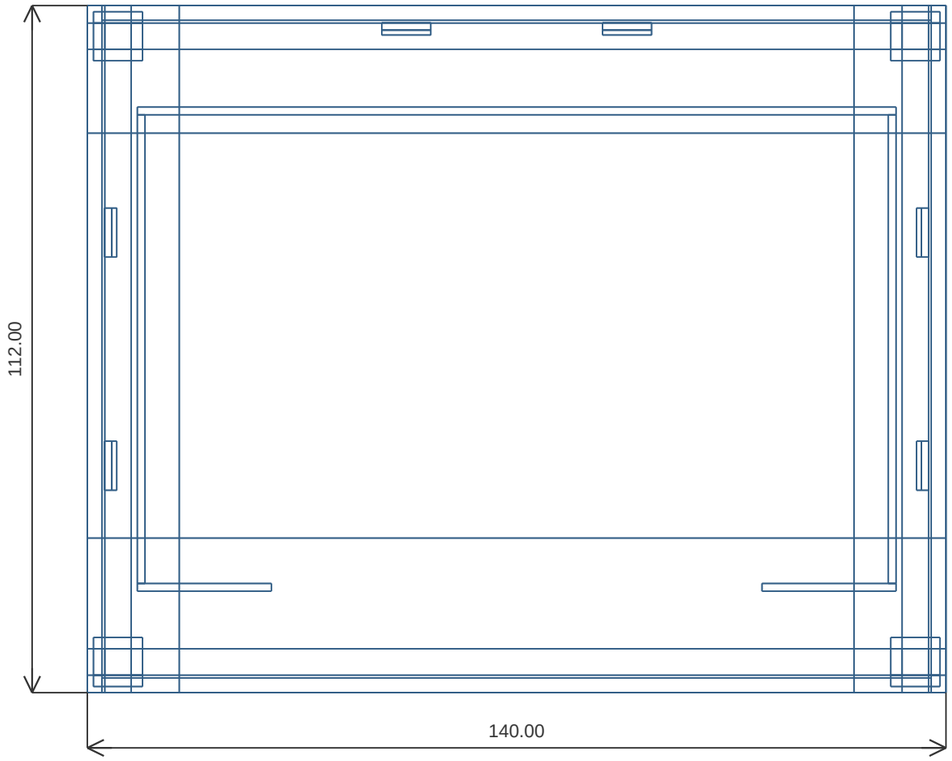
100 mm calibration bar



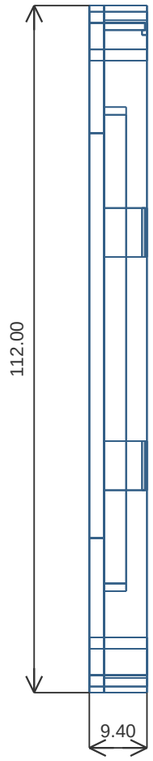
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - FRONT SHELL 5.0

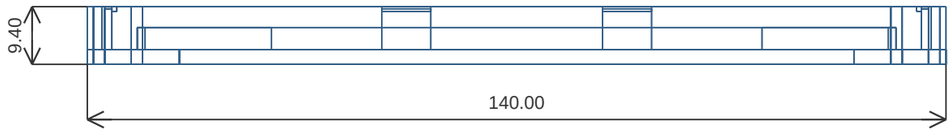
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



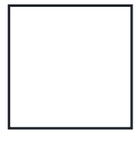
CONTROLLED GEOMETRY

- Source: front_shell_50_snap.stl
- STL bounds: 140.00 x 112.00 x 9.40
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: XY display face on bed; supports: none.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



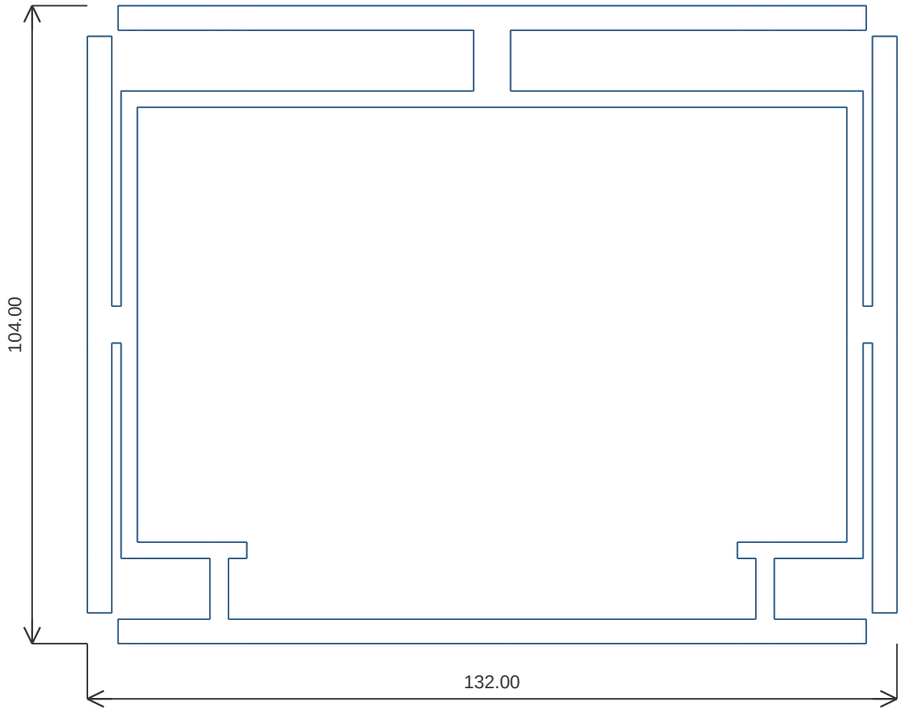
100 mm calibration bar



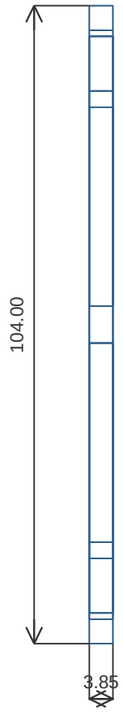
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - LCD RETAINER 5.0

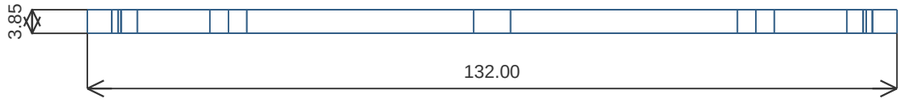
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



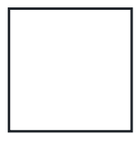
CONTROLLED GEOMETRY

- Source: lcd_retainer_50_snap.stl
- STL bounds: 132.00 x 104.00 x 3.85
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: XY flat face on bed; supports: none.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



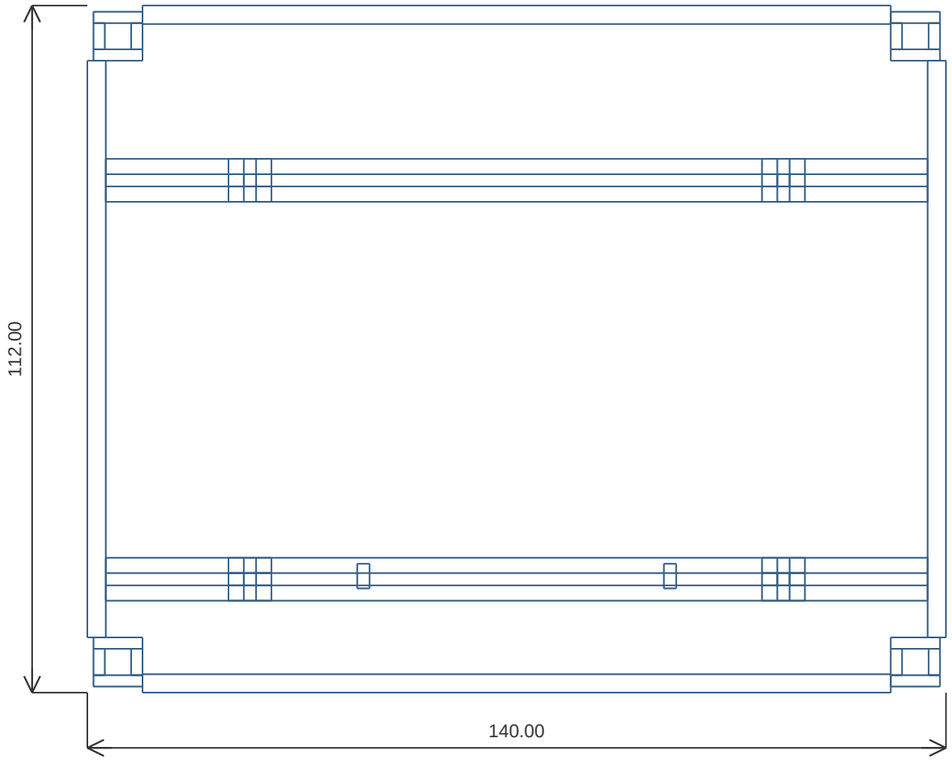
100 mm calibration bar



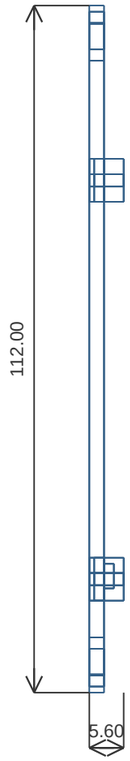
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - DOCK TRAY / M2.5

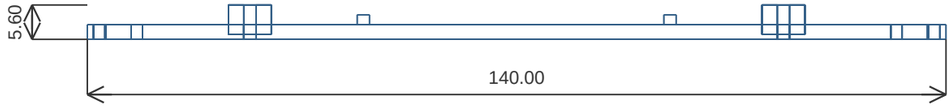
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



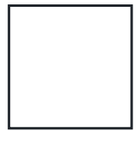
CONTROLLED GEOMETRY

- Source: dock_tray_screw_common.stl
- STL bounds: 140.00 x 112.00 x 5.60
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- 4 x official holes on 87.00 x 65.00 pitch.
- Boss 7.00 square; pilot 2.00 square x 4.80 deep.
- Hardware: 4 x M2.5 x 6 pan-head screws.
- Open FPC rails: no bridge; secure with polyimide tape.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



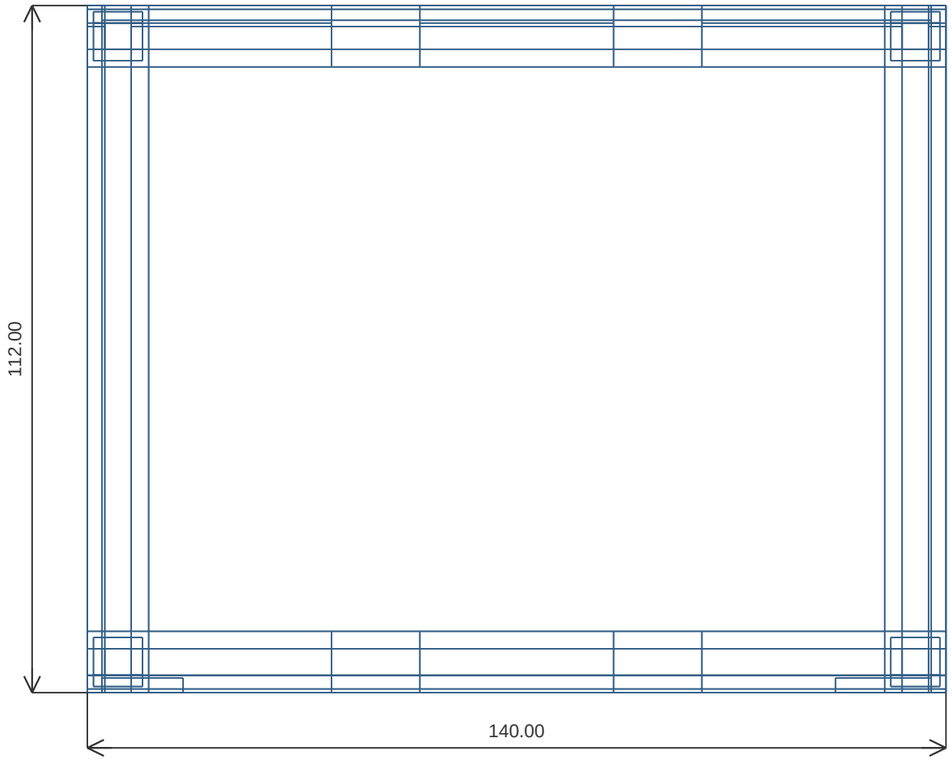
100 mm calibration bar



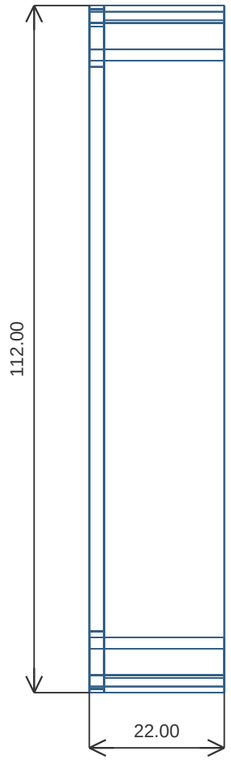
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - REAR ACCESS FRAME

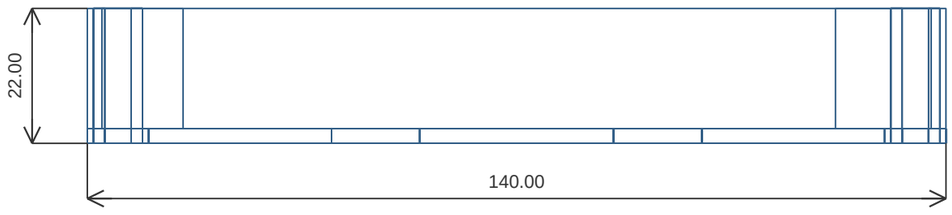
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



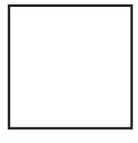
CONTROLLED GEOMETRY

- Source: rear_access_frame_common.stl
- STL bounds: 140.00 x 112.00 x 22.00
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: open flange XY face on bed; supports: none.
- Center opening is 120.00 x 92.00 for direct SODIMM/PMOD access.
- Frame remains fixed by the four original 37.00 mm case pins.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



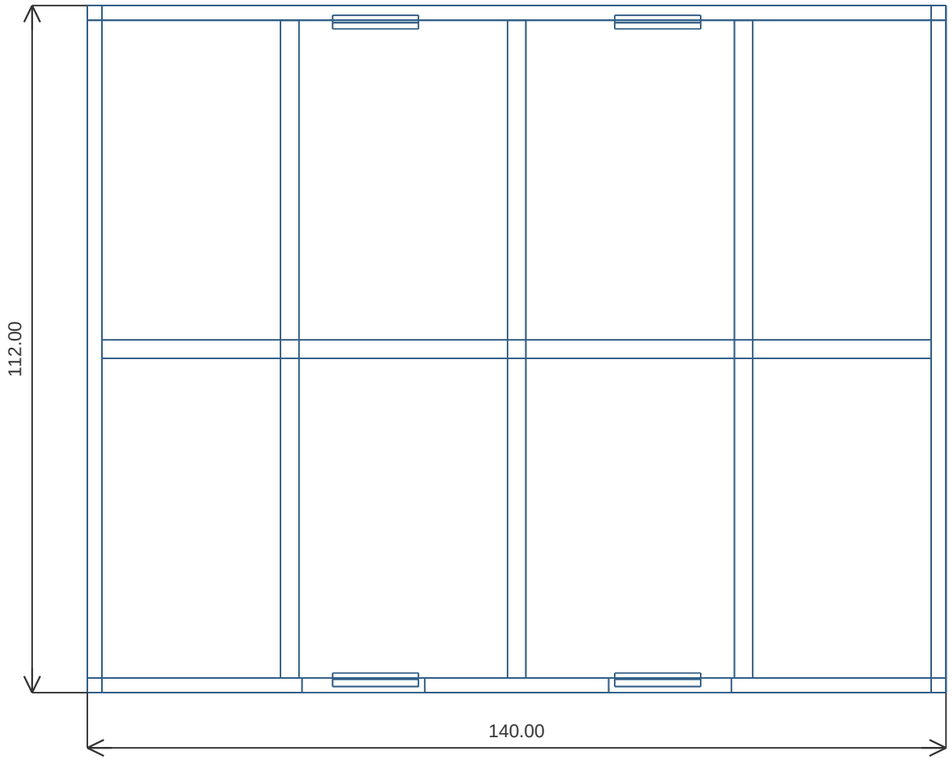
100 mm calibration bar



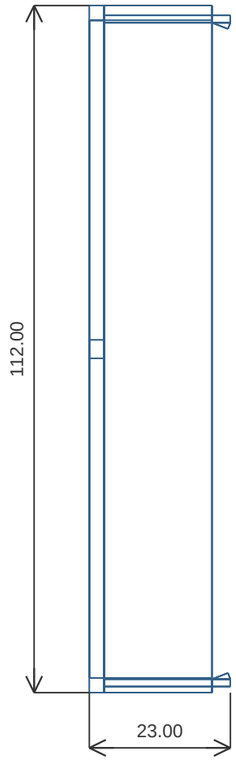
1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - REAR SERVICE CAP

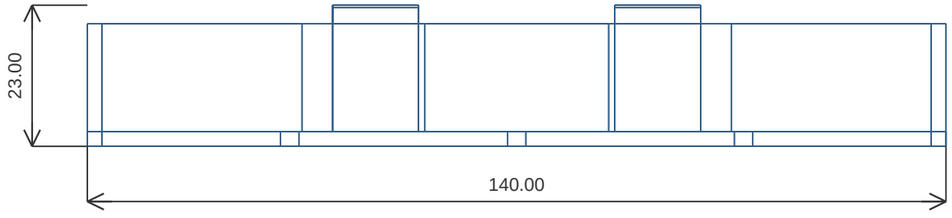
FRONT / XY - STL FEATURE EDGES



RIGHT / ZY



TOP / XZ



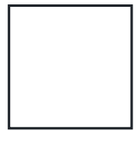
CONTROLLED GEOMETRY

- Source: rear_service_cap_common.stl
- STL bounds: 140.00 x 112.00 x 23.00
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Print orientation: rear lattice XY face on bed; supports: none.
- Install upper hooks first; rotate lower edge until both tabs click.
- Press both lower tabs inward, rotate outward and lift to remove.
- Two 20.00 mm open-edge cable exits retain connected PMOD leads.

VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

20 x 20



100 mm calibration bar



1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - CASE SNAP PIN

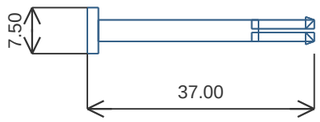
CONTROLLED GEOMETRY

- Source: case_snap_pin.stl
- STL bounds: 37.00 x 7.50 x 7.50
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Quantity: 4. Export orientation is the required flat orientation.
- Print fit_coupon.stl before the full enclosure.

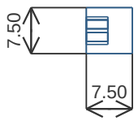
VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

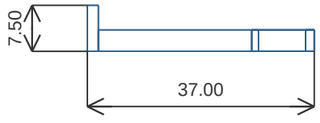
FRONT / XY - STL FEATURE EDGES



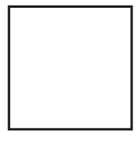
RIGHT / ZY



TOP / XZ



20 x 20



100 mm calibration bar



1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.

PRINTED PART - FIT COUPON

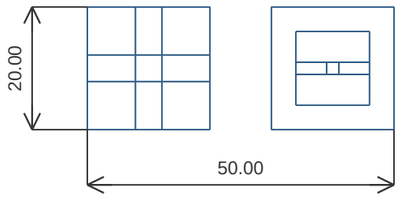
CONTROLLED GEOMETRY

- Source: fit_coupon.stl
- STL bounds: 50.00 x 20.00 x 5.60
- Views are direct orthographic feature-edge projections.
- Coplanar STL facet diagonals are suppressed only.
- Left: 4.30 square case-pin path. Right: 2.00 square M2.5 blind pilot.
- Use production material, orientation and slicer compensation.

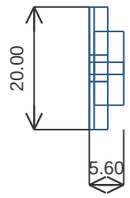
VERIFY BEFORE FINAL PRINT

- Blue lines are generated from the delivered STL, not redrawn sketches.
- Print at 100%; verify the 100 mm bar and 20 mm square.

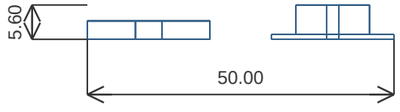
FRONT / XY - STL FEATURE EDGES



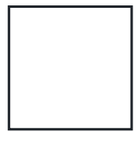
RIGHT / ZY



TOP / XZ



20 x 20



100 mm calibration bar



1:1 SHEETS: PRINT AT 100% / ACTUAL SIZE. DISABLE FIT-TO-PAGE.